

TransResUNet: Improving U-Net Architecture for Robust Lungs Segmentation in Chest X-rays

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Abstract—Medical image segmentation is regarded as an important component in a computer-aided diagnosis (CAD) system as it directly affects overall system performance. In this paper, we propose a new fully convolutional encoder-decoder model for lung segmentation named TransResUNet. We developed this architecture improving the state-of-the-art U-Net model. As part of the improvement to the classical U-Net, we introduced a pre-trained encoder, a special skip connection and a post-processing module in the proposed architecture. As a result, the proposed model outperformed the baseline U-Net model by 97.6% vs 94.9% considering the dice coefficient and 98.5% vs. 96.8% in terms of accuracy. The proposed TransResUNet achieved this feat with about 24% fewer parameters than the baseline U-Net. The implementation (based on Keras) of our proposed model is publicly available at <https://sakibreza.github.io/TransResUNet> with additional resources.

Index Terms—Chest X-ray; Deep Learning; Lungs; Segmentation.

I. INTRODUCTION

The chest X-ray (CXR) is considered a common way to diagnose a variety of diseases related to lungs and heart. One of the major applications of CXR is to assist diagnosis and inspect treatment for different lung conditions such as pneumonia, emphysema, and cancer. Concerning these sorts of diseases, a radiologist generally focuses on the lungs region for an effective interpretation of CXRs [1]. In the case of developing a computer-aided diagnosis (CAD) system, the interpretation concepts also remain the same since a diagnosis system should mimic a human radiologist. Therefore, in the development pipeline of such a CAD system, we need to include an effective lung segmentation unit.

In recent years, impressive progress has been made in the domain of deep learning and computer vision. A range of medical image segmentation models has also been developed with this advancement. U-Net [2] architecture is one of them and is regarded as a standard model for medical image segmentation. Regardless of its wide acceptability, we still found some scopes of alterations to make it more improved. In this paper, we present a new fully-convolutional model for lung segmentation called TransResUNet transforming the classical U-Net architecture.

The composition of this paper is as follows. Section II and Section III describe the baseline U-Net model and the proposed model respectively. The experimental results are

discussed in Section IV with a brief description of the data. Finally, Section V concludes the paper.

II. BASELINE U-NET MODEL

The architecture of U-Net [2] model consists of two modules - encoder and decoder. The role of the encoder is to extract spatial features of input images and the decoder part builds the mapping of segmentation using encoded features. The encoder module is quite similar to regular convolutional network architecture. Here, an input image goes through a number of blocks consisting of two padded 3×3 convolution layers with ReLU activation and max-pooling consecutively. After each max-pooling operation, the number of feature channels gets double and spatial dimensions become half of its previous state. Conversely, the decoder module up-samples the map of features, concatenates it with the corresponding decoder feature map and then conducts two 3×3 convolution operations with ReLU activation. In each up-sampling step, the decoder performs halving the number of feature channels and simultaneously their spatial dimension gets double. Thus, the original dimension of the input image is preserved and the network outputs the expected segmentation mask.

III. PROPOSED TRANSRESUNET MODEL

Focusing on the lung segmentation, we made some modifications in the classical U-Net model and proposed a new model named TransResUNet. The proposed model is demonstrated in figure 1 with all required definitions. The major components of our proposed method are described as follows.

1) *Pre-trained Encoder*: Employing a pre-trained encoder can improve the training of an encoder-decoder network [3]. Therefore, as an encoder, we introduced a trimmed portion of the VGG-16 model [4] pre-trained on the ImageNet dataset [5]. In a transfer learning task, the lower level weights have a greater impact as they are focused more on the primitive feature extraction. The weights of higher layers are rather dedicated to specialized feature extraction for the target task. Considering these facts, we took only the first 7 layers of VGG-16 keeping the architecture as compact as possible.

Here, first, the input grayscale image is converted to a 3 channel image to make it compatible with the pre-trained encoder and then it enters the main encoder. The first two encoder blocks contain two sets of 3×3 convolutions with

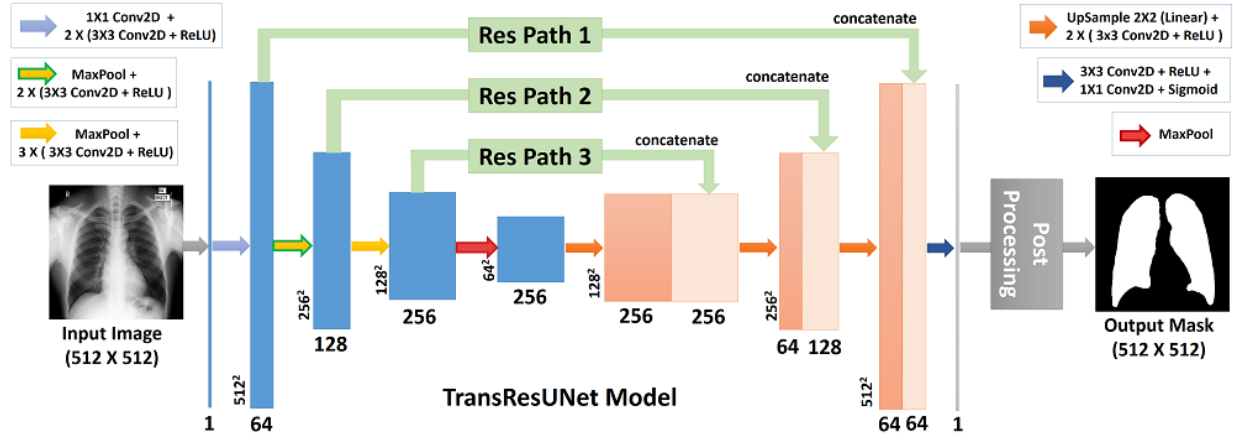


Fig. 1: Proposed TransResUNet model for lungs segmentation

ReLU activation and a max-pooling layer. The only difference in the last encoder block is that it holds three sets of 3X3 convolutions instead of two.

2) *Modified Skip Connection*: The U-Net architecture introduced us with an inventive approach to propagate spatial information from encoder to decoder which is known as skip connections. These connections generally join the encoder layers prior to max-pooling and the corresponding decoder layers after up-sampling. In spite of conserving the diffused spatial features, the traditional skip connection may cause a semantic gap that appears between the concatenated encoder and decoder features. For example, the first skip connection joins the encoder prior to the first max-pooling layer with the decoder following the last up-sampling process. In this context, the features originating from the encoder module are assumed to be comparatively primitive features since they proceed through less computational processing. On the other hand, features produced from the decoder module are assumed to be more advanced-type as they go through a lot of processing. To alleviate this semantic gap, it is supposed to be a competent idea to use a series of convolution blocks in the skip connection. Res path [6] is one of these types of connection which can solve this situation. Figure 2 shows how we constructed a res path using a number of residual blocks.

3) *Decoder*: The decoder module comprises three decoder blocks. Each of these blocks consists of an up-sampling that follows two 3X3 convolution layers with ReLU activation. Here, we used the nearest neighbor up-sampling as it is effective in limiting artifacts at the edge of the predicted output form. Again, features of the last convolution layer in a decoder block are concatenated with the corresponding encoded features from its residual connection. Finally, the last layer of the decoder produces the predicted output mask followed by a 1X1 convolution and sigmoid activation.

4) *Post-processing*: While experimenting with our model, we encountered that a few predicted masks have defects like-holes in the mask, overlapping left and right masks, and unexpected objects out of the real mask region. To reduce these output noises, we added a post-processing segment at

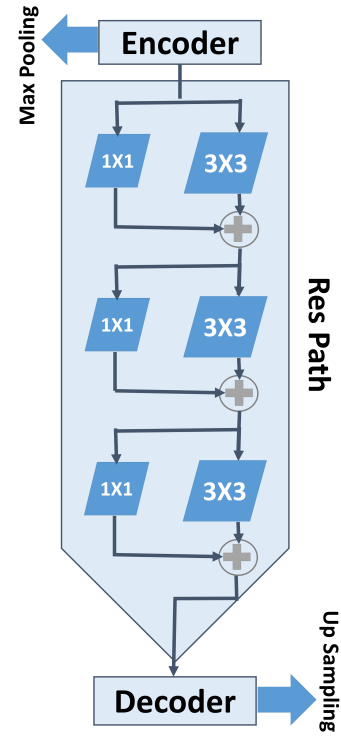


Fig. 2: Res path contains blocks of two parallel 3X3 and 1X1 convolution filter and the addition of their outputs. In proposed model, the first, second and third res path (according to figure 1) contain 3, 2 and 1 such block(s) respectively.

the end of our lung segmentation pipeline. The proposed post-processing steps (figure 3) are performed as follows.

First, we applied the flood-fill algorithm for filling up the undesired holes in the mask. The algorithm is performed by choosing the top left corner pixel (0,0) of the mask as the starting node. Here, the target color was black(0) like the starting node and the replacement color was white(1). The resultant image was inverted and unioned (bitwise OR) with the main image to get the desired output similar to figure 3(a).

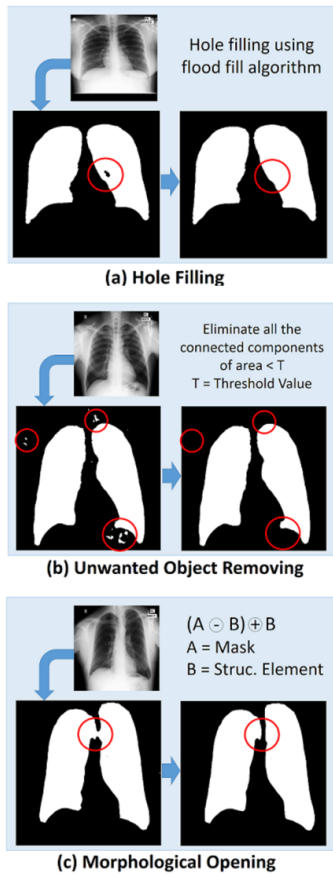


Fig. 3: Visualization of the post-processing steps

Secondly, there can remain some undesired objects around the lung masks. To remove them, we determined all the connected components in the mask and eliminated all the components with an area less than a predefined threshold value. Finally, we performed a morphological opening on the predicated mask using an elliptical structuring element. Figure 3(c) visualizes how opening detaches the left and right masks on an overlapped mask sample.

IV. EXPERIMENTAL RESULTS

A. Data

For our experimental purpose, we used the Montgomery Country (MC) dataset [7]. This dataset is managed by the National Library of Medicine (NLM) in collaboration with the Department of Health and Human Services of Montgomery Country, Maryland. It consists of 138 posterior-anterior chest X-Ray images including 80 normal and 58 abnormal X-Rays. These images are provided with corresponding manual lung masks that were annotated with the support of professional radiologists. For our purpose, we resized all the images to the pixel resolution of 512×512 .

B. Evaluation Metrics

In model evaluation, we applied two widely used metrics for image segmentation tasks - dice similarity coefficient and

pixel accuracy. The definitions of these metrics are as follows.

a) *Dice Similarity Coefficient*: The dice coefficient is a standard metric for evaluating segmentation performance. It is defined as twice of the overlapped area between the predicted mask and the ground truth mask divided by the total area sum of the predicted mask and the ground truth mask. Equation 1 shows how this metric is calculated.

$$Dice = \frac{2|PM \cap GM|}{|PM| + |GM|} = \frac{2|TP|}{2|TP| + |FN| + |FP|} \quad (1)$$

Here, PM = predicted mask and GM = Ground Truth Mask.

b) *Pixel Accuracy*: The pixel accuracy is the simplest way to measure segmentation performance. Pixel accuracy is defined as the sum of the number of true positive and true negative pixels divided by the total number of pixels. Equation 2 presents the process of calculating the pixel accuracy.

$$Accuracy = \frac{|TP| + |TN|}{|TP| + |TN| + |FP| + |FN|} \quad (2)$$

Here, TP = True Positives, TN = True Negatives, FP = False Positives and FN = False Negatives.

C. Model Evaluation

For evaluation of our proposed method, we randomly split our dataset into three segments - train set (70%), validation set(10%), and test set (20%). Then, we trained the baseline U-Net model and the proposed TransResUNet model for 150 epochs on a Tesla K80 GPU with data augmentation.

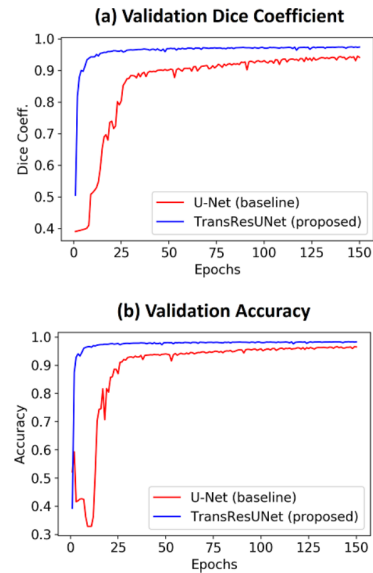


Fig. 4: Progress comparison of validation performance with the number of training epochs.

In the training phase, we observed the validation progress of both the baseline and the proposed model. Fig 4 shows the comparison of validation progress between two models. From this, we notice that the proposed model with a pre-trained

encoder reached a fair state with a minimal number of epochs. Eventually, it outperforms the U-Net model in both metrics - dice similarity coefficient and accuracy (binary accuracy).

In addition to comparing the proposed model with U-Net, we set up two variants of the proposed model - one without pre-trained weight and another without res path for further justification. Table I demonstrates test results comparison among the baseline U-Net, variants of TransResUNet and finally the proposed TransResUNet model. Here, we see that the proposed TransResUNet outperformed the baseline U-Net model with 24.4% fewer parameters. Furthermore, the proposed model is proved better than its variants which validates the addition of pre-trained encoder and res path. For a simpler interpretation of the result, we visualize some sample predictions of our model comparing with the predictions of baseline U-Net and corresponding ground truths in figure 5.

TABLE I
SEGMENTATION RESULTS COMPARISON AMONG THE
BASELINE U-NET MODEL, VARIANTS OF TRANSRESUNET
AND THE PROPOSED TRANSRESUNET MODEL

Model	Param.*	Dice (%)	Acc. (%)
U-Net (baseline)	7.8M	94.9	96.8
TransResUNet (random initialization)	5.9M	92.1	95.3
TransResUNet (without res. path)	5.8M	94.0	96.9
TransResUNet (proposed)	5.9M	97.6	98.5

*Parameters in Millions (M).

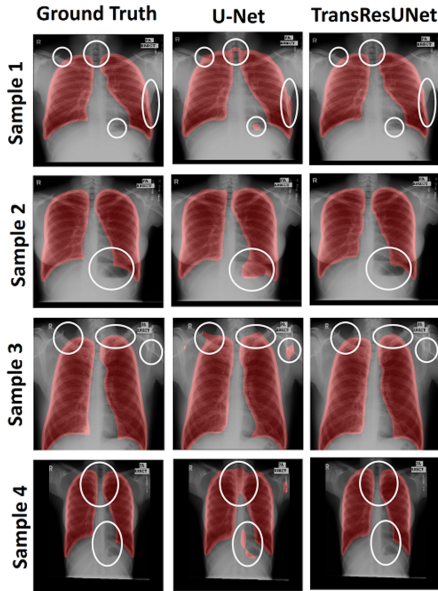


Fig. 5: Comparison of sample segmentation predictions. Here, masks are shown in the form of color maps over input images.

Finally, Table II shows the comparison among different studies conducted on the same dataset. From other studies, S. Candemir et. al. [7] proposed an approach which involves anatomical atlases with non-rigid registration. Apart from this, P. Chondro et. al. [8] introduced a low order region growing method to segment the lungs. The most recognized study

on the lungs segmentation task is conducted by et. al. [9]. They used state-of-the-art U-Net architecture which is the baseline model in our study. This comparison further verifies the robustness of our model on lung segmentation.

TABLE II
COMPARISON OF LUNGS SEGMENTATION STUDIES ON MC
DATASET

Approach	Dice (%)	Acc. (%)
Anatomical atlases with Non-rigid Registration [7]	-	94.1
Low order adaptive region growing [8]	-	96.6
U-Net [9]	95.4	97.7
TransResUNet (proposed)	97.6	98.5

V. CONCLUSION

The results show that our segmentation model improved the effectivity of classical U-Net with less number of parameters. One of the major limitations of using such fully convolutional model is that they need a large number of training data. In our model, we have solved this problem by using a transfer learning approach. Furthermore, we alleviated a semantic gap issue in the skip connection by replacing it with a residual connection. As a future plan, we would like to evaluate the TransResUNet model with other medical image datasets, modify it further and transform it into a generic model for medical image segmentation.

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